

DATABASE MANAGEMENT SYSTEMS LAB

LAB 3: ER MODEL AND RELATIONAL SCHEMA

Question 1: Manufacturer

(a) ER Model

The entities considered are PRODUCT, COMPONENT and SUPPLIER.

1. PRODUCT

Attributes: Product_ID, Product_Name, Quantity_On_Hand

Primary Key: Product_ID

2. COMPONENT

Attributes: Component_ID, Component_Name, Description

Primary Key: Component_ID

3. SUPPLIER

Attributes: Supplier_ID, Supplier_Name

Primary Key: Supplier_ID

Relationships:

PRODUCT — MADE_UP_OF — COMPONENT

Cardinality: Many-to-Many (M:N)

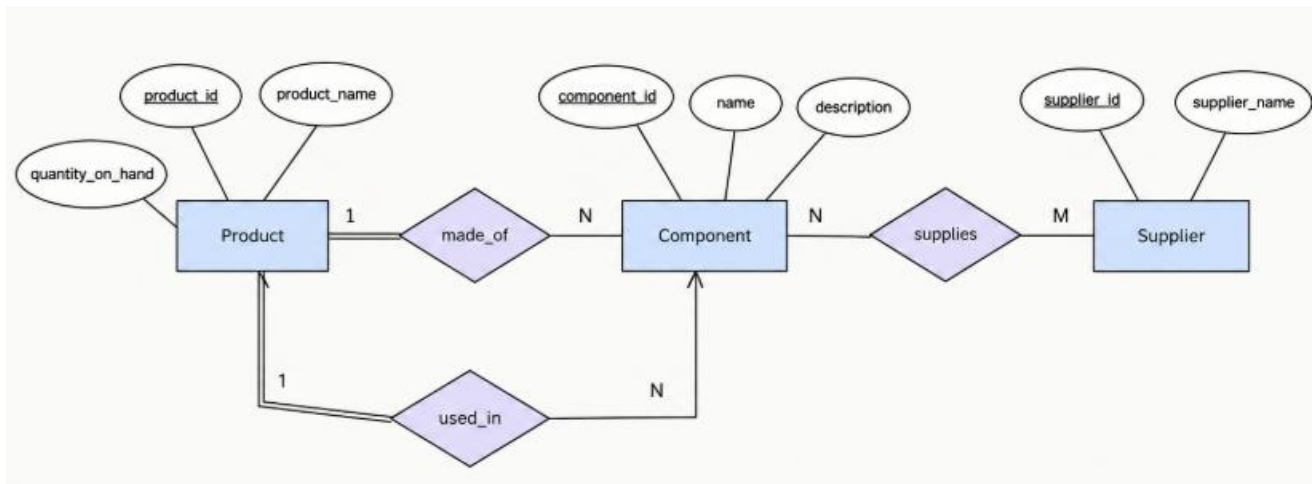
Participation: PRODUCT has total participation because every product must have at least one component. COMPONENT has partial participation because a component may not be used in any product.

SUPPLIER — SUPPLIES — COMPONENT

Cardinality: Many-to-Many (M:N)

Participation: Both SUPPLIER and COMPONENT have partial participation. A supplier can exist without supplying a component and a component can exist without a supplier.

Simple ER representation:



(b) Relational Schema

PRODUCT(Product_ID PK, Product_Name, Quantity_On_Hand)

COMPONENT(Component_ID PK, Component_Name, Description)

SUPPLIER(Supplier_ID PK, Supplier_Name)

PRODUCT_COMPONENT(Product_ID PK, FK, Component_ID PK, FK)

SUPPLIES(Supplier_ID PK, FK, Component_ID PK, FK)

PRODUCT_COMPONENT is used for the many-to-many relationship between products and components. SUPPLIES is used for the many-to-many relationship between suppliers and components.

Assumptions

- Product_ID, Component_ID and Supplier_ID uniquely identify their respective entities.
- A product must contain at least one component.
- A component may be unused and may have no supplier.
- A supplier may exist even if the supplier is not currently supplying any component.

Question 2: Car Dealership

(a) ER Model

The entities considered are CAR, CUSTOMER, SALESPERSON, SALE, INVOICE, SERVICE_TICKET, MECHANIC and PART.

1. CAR

Attributes: Serial_Number, Make, Model, Year, Condition

Primary Key: Serial_Number

2. CUSTOMER

Attributes: Customer_ID, Customer_Name, Phone, Address

Primary Key: Customer_ID

3. SALESPERSON

Attributes: Salesperson_ID, Salesperson_Name, Phone

Primary Key: Salesperson_ID

4. SALE

Attributes: Sale_ID, Sale_Date, Sale_Price

Primary Key: Sale_ID

5. INVOICE

Attributes: Invoice_ID, Invoice_Date, Total_Amount

Primary Key: Invoice_ID

6. SERVICE_TICKET

Attributes: Ticket_ID, Service_Date, Service_Description, Service_Status

Primary Key: Ticket_ID

7. MECHANIC

Attributes: Mechanic_ID, Mechanic_Name, Phone

Primary Key: Mechanic_ID

8. PART

Attributes: Part_ID, Part_Name, Unit_Price

Primary Key: Part_ID

Relationships:

CAR — IS_SOLD_IN — SALE

Cardinality: One car can be unsold or sold once; every sale is for one car.

CUSTOMER — BUYS — SALE

Cardinality: One customer can buy many cars; every sale has one customer.

SALESPERSON — HANDLED_BY — SALE

Cardinality: One salesperson can handle many sales; every sale has one salesperson.

SALE — HAS_INVOICE — INVOICE

Cardinality: One-to-One. Exactly one invoice is created for each sale.

CAR — HAS_SERVICE_TICKET — SERVICE_TICKET

Cardinality: One car can have many service tickets; every service ticket is for one car.

CUSTOMER — REQUESTED_BY — SERVICE_TICKET

Cardinality: One customer can have many service tickets. A customer may use service without buying a car.

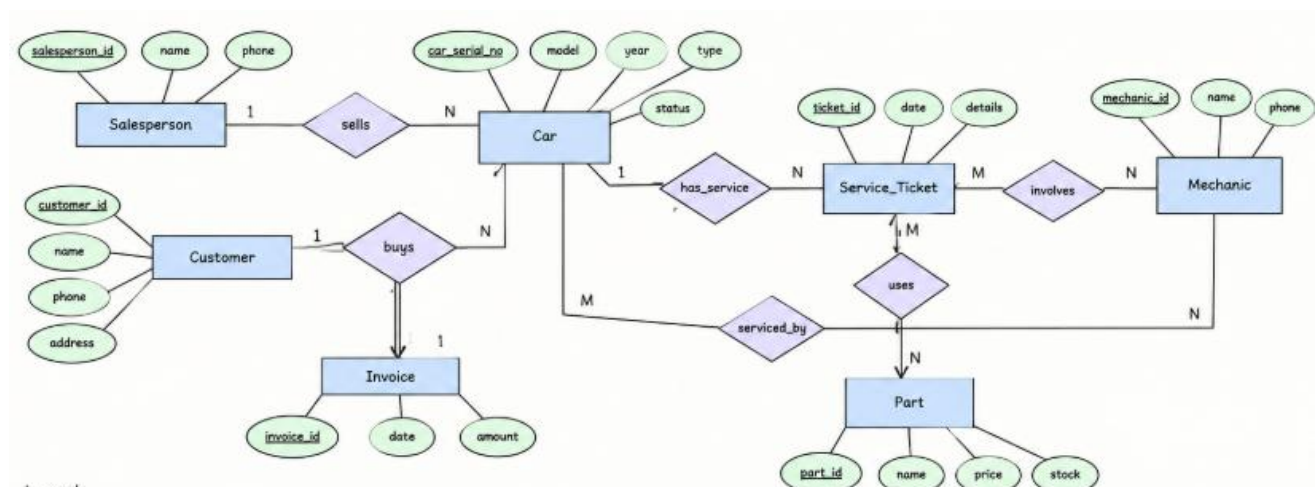
MECHANIC — WORKS_ON — SERVICE_TICKET

Cardinality: Many-to-Many. A service ticket may involve many mechanics and a mechanic may work on many service tickets.

SERVICE_TICKET — USES — PART

Cardinality: Many-to-Many. A service ticket may use zero, one or several replacement parts.

Simple ER representation:



(b) Relational Schema

CAR(Serial_Number PK, Make, Model, Year, Condition)

CUSTOMER(Customer_ID PK, Customer_Name, Phone, Address)

SALESPERSON(Salesperson_ID PK, Salesperson_Name, Phone)

SALE(Sale_ID PK, Sale_Date, Sale_Price, Serial_Number FK, Customer_ID FK, Salesperson_ID FK)

INVOICE(Invoice_ID PK, Invoice_Date, Total_Amount, Sale_ID FK)

SERVICE_TICKET(Ticket_ID PK, Service_Date, Service_Description, Service_Status, Serial_Number FK, Customer_ID FK)

MECHANIC(Mechanic_ID PK, Mechanic_Name, Phone)

PART(Part_ID PK, Part_Name, Unit_Price)

SERVICE_MECHANIC(Ticket_ID PK, FK, Mechanic_ID PK, FK)

SERVICE_PART(Ticket_ID PK, FK, Part_ID PK, FK)

SERVICE_MECHANIC is used to represent the many-to-many relationship between mechanics and service tickets. SERVICE_PART is used to represent the many-to-many relationship between service tickets and parts.

Assumptions

- Serial_Number uniquely identifies a car and is used for maintaining service history.
- A car can exist in the dealership without being sold.
- A customer can bring a car for service without purchasing a car from the dealership.
- Each service ticket belongs to one car and one customer.
- No extra attributes such as tax, discount or payment method are added because they are not specified in the question.